Python Lab exercise- 2

1. A triangle has 3 sides a,b,c as 17,23,30. Calculate and display the area using the formula s=(a+b+c)/2 and area =squareroot(s(s-a)(s-b)(s-c)) using the math module
2. Write a program to repeat the string “Python” n times. n should be entered by user.
3. Write a program that reads a number of seconds and prints it in minutes and seconds
4. Write a program to take year as input and check if it is a leap year or not
5. Write a program to take 2 numbers and print if the first number is fully divisible by the second number or not.
6. Write a program to input a number and print the reversed number
7. Write a program to take inputs for day and month and then calculate which day of the year the given date is. (Take 30 days for all months) For eg.If input is day 3 and month 2 , then it should print” Day of the year 33”.
8. Write a program to input a string and then prints it equal to number of times its length
9. Write a program to input a number and check whether it is odd or even
10. Write a program to emulate the working of a calculator using if-elif
11. Write a program to find the greatest of 3 numbers using nested if
12. Write a program to calculate the roots of a quadratic equation
13. Write a program to print the sum of the first 20 natural numbers
14. Write a program to input a number and find the factorial of a number
15. Write a program to read m and n and find m raised to n(Do not use exponentiation operator or math module)
16. Write a program to calculate the sum of even and odd numbers of the first n natural numbers where n is read from the keyboard
17. Write a program to check whether a number is prime or not
18. Write a program to generate the divisors of a number
19. Write a program to check whether a number is perfect or not
20. Write a program to check whether a number is palindrome or not
21. Write a program to print the multiplication table of a number. The number has to be entered by the user
22. Write a program that has the options to accept the marks of the student in 5 subjects and display the same. Display the percentage and find the grade of the student as per the following criteria

Critera	Grade
Percentage > 85	A
Percentage < 85 && percentage >=75	B
Percentage < 75 && percentage >=50	C
Percentage < 50 && percentage >=30	D
Percentage < 30	Fail

23. Write a Python program to print the following shapes

a)
```
*
* *
* * *
* *
*
```

b)
```
0
2 2
4 4 4
6 6 6 6
8 8 8 8 8
```